



# Image-to-Image translation

Computer Vision II

Master in Artificial Intelligence

2025/2026

## 1 Overview

The goal of this project is the use of **image-to-image translation models**, such as Pix2Pix [1] to **generate realistic images from input label maps** in diverse application domains. Pix2Pix employs adversarial training to learn a mapping function from an input image domain to an output image domain. This process involves simultaneously training a generator network to produce realistic images and a discriminator network to distinguish between generated and real images.

This project should be carried out in **teams of two students**, encouraging collaboration, problem-solving, and shared responsibility in the development process.

## 2 Materials

Six datasets from different application domains are provided for this project. Each team will be **randomly assigned a dataset**.

Table 1 summarizes the general information and Figure 1 displays an example of image from each dataset. Each image in the datasets is a paired sample composed of a real image (left side) and its corresponding label map (right side).

## 3 Methods

The objective of this assignment is the development of an image-to-image translation method that, given a label map provides a realistic image.

### Task I: Baseline approach

1. Design and implement a baseline version of the Pix2Pix model.
2. Train the model on the assigned dataset (use training, validation, and testing splits).
3. Compute performance metrics.

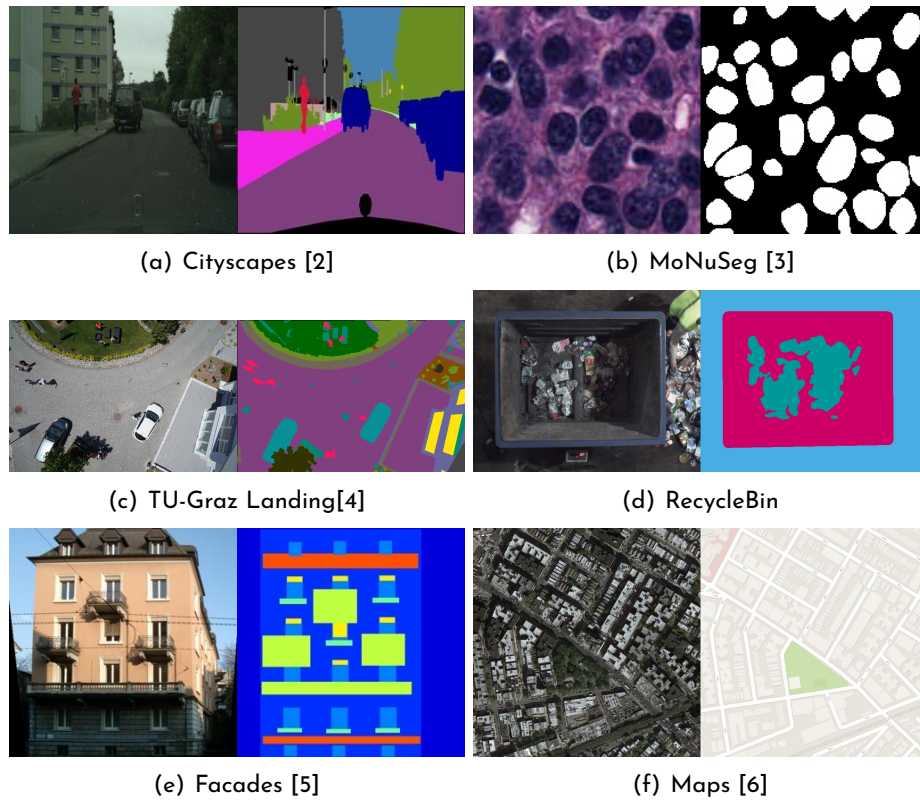


Figure 1: Samples of paired data from each dataset where the left side displays the real image and the right side its corresponding label map. (a) Semantic labels for urban street scenes. (b) Nuclei cell labels for organ tissues. (c) Aerial semantic labels for safe UAV Landing. (d) Occupancy labels for recycling containers (e) Architectural labels for build facades. (f) Geographic labels for satellite images.



Table 1: General information for each dataset

| Dataset         | Number of images | Size             |
|-----------------|------------------|------------------|
| Cityscapes      | 500              | $256 \times 256$ |
| MoNuSeg         | 500              | $200 \times 200$ |
| TU-Graz Landing | 400              | $400 \times 600$ |
| RecycleBin      | 217              | $480 \times 600$ |
| Facades         | 500              | $256 \times 256$ |
| Maps            | 500              | $600 \times 600$ |

### Task 2: Improvements

1. Propose potential improvements or select at least one of the following:
  - Explore different architectures
  - Analyze other loss functions
  - Apply data augmentation
  - Use transfer learning
2. Train the model
3. Compute performance metrics and compare with Task 1

### Task 3: Demo

1. Write a small program that receives an image as input and makes inference using the trained models
2. Create a public Github repository to upload your demo program
  - The program to make **only** inference
  - One or several trained models
  - Some sample images
  - README with instructions for installing and testing the demo



## 4 Submission

The **due date** for this project is **May 24th, 2026**.

Each team should upload in the corresponding Teams assignment the following items:

- The code written in Python
- A report describing and justifying the proposed approach. Use the IEEE conference template<sup>1</sup> to generate a PDF. The length of the report is limited to 4 pages. A rubric is also provided in order to provide guidance in the writing. The following is a suggested structure for the report:
  - Title, authors
  - Abstract
  - Introduction to the problem and the proposed approach.
  - Technical approach (include the link to the public repository)
  - Results
  - Conclusion
  - References

Each team will also have to discuss their approach in a Teams meeting with an instructor after the submission.

## 5 Peer review

After the submission date, each team will be required to review some peer's works and assign a score that is **clearly justified** with specific comments linked to the provided rubric.

The **due date** for the reviewing process is **May 31st, 2026**.

Failure to complete the peer-review task or to provide a well-reasoned justification may result in a reduction of the overall score.

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<sup>1</sup><https://www.ieee.org/conferences/publishing/templates.html>



## 6 Assessment

The final mark for the assignment (60% of the total score) will be determined in the following way:

- 50% will be awarded by the teacher
- 10% will be derived from peer review

## References

- [1] Phillip Isola, Jun-Yan Zhu, Tinghui Zhou, and Alexei A Efros. Image-to-image translation with conditional adversarial networks. CVPR, 2017.
- [2] Marius Cordts, Mohamed Omran, Sebastian Ramos, Timo Rehfeld, Markus Enzweiler, Rodrigo Benenson, Uwe Franke, Stefan Roth, and Bernt Schiele. The cityscapes dataset for semantic urban scene understanding. In Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016.
- [3] Neeraj Kumar, Ruchika Verma, Sanuj Sharma, S. K. Bhargava, Abhishek Vahadane, and Amit Sethi. A dataset and a technique for generalized nuclear segmentation for computational pathology. IEEE Transactions on Medical Imaging, 36:1550-1560, 2017.
- [4] Safe-UAV-Landing. <https://www.tugraz.at/index.php?id=22387>, 2021. [Last accessed on March 14th, 2025].
- [5] Radim Šára, Radim Tyleček. Spatial pattern templates for recognition of objects with regular structure. In Proc. GCPR, Saarbrücken, Germany, 2013.
- [6] Pix2Pix Datasets. <http://efrosgans.eecs.berkeley.edu/pix2pix/datasets/>, 2018. [Last accessed on March 14th, 2025].